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Mini Project Synopsis

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Batch No: 38	Guide Name: Dr.Usha B.A	Submission Date: 18-09-2026
Mini Project Title : IPOLENS: Machine Learning for Smarter IPO Insights & Investment Tracking		
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Abstract

IPOLENS is a unified web platform that binds the fragmented Indian IPO ecosystem into a single application, bringing together live IPO tracking, AI-assisted prospectus analysis, family multi-PAN application management, one-click allotment verification, and secure demat portfolio tracking. Retail investors today rely on multiple disconnected sources – news portals for Grey Market Premium (GMP), exchange websites for subscription data, registrar portals for allotment results, and separate broker apps for portfolio tracking – which leads to fragmented information, delayed decisions, and error-prone manual record keeping, especially when applying through several family PAN accounts. IPOLENS addresses this by providing a live IPO radar covering Mainboard and SME issues with real-time GMP, category-wise subscription telemetry (Retail, HNI/NII, QIB), and interactive listing-performance charts. An AI-powered investment analyst module summarizes lengthy Draft Red Herring Prospectus (DRHP) filings and produces a Subscribe / Avoid / Neutral verdict along with supporting strengths and risks, reducing the effort needed to interpret complex regulatory documents. A Family Multi-PAN Manager allows investors to record applications, lots, and blocked amounts across relationship-tagged PAN profiles while validating against duplicate bids that are disallowed under SEBI regulations. The platform further offers one-click deep links to registrars such as Link Intime, KFin Technologies, and Bigshare Services, and a result scanner that updates allotment status automatically. Finally, a password-free portfolio tracker imports CSV and CAS statements from brokers such as Groww, Zerodha Kite, and 5paisa, or directly from NSDL/CDSL, matching ISIN codes to compute real-time profit and loss without sharing user broker login credentials. Together, these modules aim to provide a single secure dashboard for retail investors' complete IPO lifecycle, from discovery through post-listing portfolio management.

Introduction

The Indian primary market has witnessed a sharp rise in retail participation over the last few years due to an increase in demat account openings, growing financial awareness, and easy access to trading applications. However, the tools a retail investor has to research and act on an IPO are scattered across many independent platforms. IPOLENS is proposed as a single, integrated web-based dashboard bringing together IPO discovery, AI-driven prospectus analysis, multi-member family application tracking, allotment verification, and demat portfolio management under one roof.

The platform is designed around five core modules - a Live IPO Radar, an AI-Powered Investment Analyst, a Family Multi-PAN Manager, a 1-Click Allotment Checker and a Safe Demat Portfolio Tracker - each targeting a specific pain point in the existing IPO investment workflow while ensuring sensitive information such as broker credentials never has to be shared with the platform.

Motivation

Retail investors routinely apply for a single IPO from multiple family PAN accounts (self, spouse, parents) to increase their probability of allotment but this is presently tracked manually on paper or in spreadsheets which is error-prone and does not scale. On allotment day, registrar websites such as those of Link Intime and KFin Technologies frequently slow down or crash under heavy traffic and investors are forced to re-enter PAN and application details several times through broken captchas. Separately, understanding if an IPO is worth subscribing to requires reading a Draft Red Herring Prospectus that can run into several hundred pages of legal and financial disclosure which is impractical for most retail investors to do for every upcoming issue. Finally, tracking how allotted IPO shares perform after listing along with the rest of one's portfolio typically requires linking a third party application to a broker account and sharing login credentials, a privacy and security concern. These recurring, first-hand frustrations motivated the design of a single platform to avoid visiting five or six different websites and spreadsheets for every IPO application cycle.

Problem Statement

Existing solutions only partially address the IPO investment lifecycle. IPO-tracking portals such as Chittorgarh and IPO Central provide GMP and subscription data but not allotment automation, family-level application tracking or portfolio integration. Broker applications such as Zerodha Kite, Groww and 5paisa allow a user to apply for an IPO and view their own portfolio but do not consolidate applications made from other family members' accounts held with different brokers and do not summarize prospectus documents. Registrar websites (Link Intime, KFin Technologies, Bigshare Services) provide authoritative allotment data but require the investor to know which registrar handled a specific issue and the portals are prone to downtime and repeated data entry on allotment days. Third party portfolio trackers who do provide a consolidated view require the investor to link their broker account through login credentials or screen-scraping which is a data privacy risk many investors do not want to accept. Consequently, there is a need for a platform that unifies IPO discovery, AI-assisted research, family application management, allotment verification and credential-free portfolio tracking within a single interface.

Objectives

- Design and implement a Live IPO Radar that tracks Mainboard and SME IPOs across Upcoming, Open, Closed and Listed stages, displaying GMP and category-wise (Retail, HNI/NII, QIB) subscription data refreshed at a defined interval, and validated against at least two independent public data sources for accuracy.
- Develop an AI-Powered Investment Analyst module that ingests a DRHP document and, in a target processing time of under one minute per document, generates a Subscribe / Avoid / Neutral verdict and the top three strengths and top three risk factors identified from the filing.
- Build a Family Multi-PAN Manager that stores relationship tagged PAN profiles (Self, Spouse, Parent, Child) for at least four family members per user account and enforces rules that block duplicate applications under the same PAN for a given issue as per SEBI regulations.

- Implement a 1-Click Allotment Checking module with deep links to at least three official registrars (Link Intime, KFin Technologies, Bigshare Services) and an automated result scanner that correctly classifies allotment status (Allotted / Not Allotted) from pasted registrar output with a target accuracy of at least 95% on a test set of sample results.
- Develop a Safe Demat Portfolio and P&L Tracker that can import CSV/CAS statements from at least three sources (e.g., Groww, Zerodha Kite, NSDL/CDSL) without requiring broker login credentials, match stock ISIN codes to company symbols with a target accuracy of at least 95%, and remove duplicate transaction rows.
- Evaluate the usability of the integrated platform through structured testing with at least ten representative users, measuring task completion time for a full IPO application-to-allotment-check workflow against the current multi-website workflow.

SDG (Sustainable Development Goal)

The project is mapped to Sustainable Development Goal 8: Decent Work and Economic Growth and specifically to Target 8.10, which calls for boosting the capacity of domestic financial institutions to encourage and expand access to banking, insurance and financial services for all.

IPOLENS supports this target by lowering the informational and procedural barriers that keep first-time and small retail investors from participating confidently in the primary capital market. By consolidating scattered IPO information, automating error prone manual processes such as family PAN tracking and allotment verification, and offering an AI generated plain-language summary of complex prospectus documents, the platform makes primary market investing more transparent and accessible to a wider base of retail investors, thus, contributing to broader financial inclusion and informed participation in the formal financial system.

Expected Outcome

- A working prototype of the IPOLENS web platform implementing the five core modules described above: Live IPO Radar, AI-Powered Investment Analyst, Family Multi-PAN Manager, 1-Click Allotment Checker and Safe Demat Portfolio Tracker.
- A documented evaluation of the AI-based prospectus summarisation and verdict generation module, benchmarked against manually prepared summaries for a sample set of recent DRHP filings.
- A usability study comparing task completion time and error rate on IPOLENS against the existing multi-website workflow for the same set of tasks.
- A technical report / conference-style paper describing the system architecture, ISIN-matching and duplicate detection algorithms, and allotment-status scanning approach, suitable for submission to a student or departmental technical symposium.
- A reusable, credential-free CSV/CAS parsing module for demat statements that can be extended to additional brokers and depositories in

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